OVER A NOETHERIAN LOCAL RING OF THE BASE, PURITY ALONG A FIBRE COINCIDES WITH UNIVERSAL PURITY

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Allowed standard prerequisites (editorial).

- The impurity lemma over a Noetherian base, restricting impurities to quasi-finite points. Same chapter flat.tex, label lemma-Noetherian-impurity-quasi-finite (invoked at proof line 4939). It controls impurities in the Noetherian situation, which is what allows the local condition to be tested on the whole fibre.
- The characterization of purity by the non existence of impurities, item (2). Same chapter flat.tex, discussion following Definition \ref{definition-pure} (invoked at proof line 4941). It converts the statement about impurities into the equivalence between purity and universal purity along the fibre.

Source reference. Original definition 05J4: sources/stacks-flat/pinned-original-source.tex, source lines 4808–4808. Navigation pointer for \ref{definition-pure} used inside the counted span (line 4941); the argument that consumes it is retained in the span.

Source reference. Original lemma 05J1: sources/stacks-flat/pinned-original-source.tex, source lines 4699–4699. Navigation pointer for \ref{lemma-Noetherian-impurity-quasi-finite} used inside the counted span (line 4939); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes which is of finite type. Let $\mathcal{F}$ be a finite type quasi-coherent $\mathcal{O}_X$-module. Let $s \in S$. If $\mathcal{O}_{S,s}$ is Noetherian then $\mathcal{F}$ is pure along X_s if and only if $\mathcal{F}$ is universally pure along X_s.*

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Proof. First we may replace S by $\mathrm{Spec}(\mathcal{O}_{S,s})$, i.e., we may assume that S is Noetherian. Next, use Lemma 05J1 and characterization (2) of purity given in discussion following Definition 05J4 to conclude. □